GA3 answers -

Part A: Your README, data, and Git repo

1: Create subdirs and copy files

2: Initialize a Git repository, and commit the README and `.gitignore`

3: Commit throughout the assignment

Part B: Basic shell scripts

4: Predict the output of `echo.sh`

Answer (prediction and reasoning for `bash scripts/echo.sh Oct07 Oct08`): Its going to give an error because we have three arguments and it only provides two *Answer* (prediction and reasoning for `bash scripts/echo.sh Oct07 Oct08 Oct09 Oct10`): Its going to give an error because we have three arguments and it provides four *Answer* (prediction and reasoning for `bash scripts/echo.sh Oct07 Oct08 "Oct09 Oct10"`): I think this would work, because it provides three arguments treating "Oct09 Oct10" as one *Answer* (prediction and reasoning for `bash scripts/echo.sh results/fastqc/*.html`): I dont think itll work because the file path is a single argument due to it being one html and its saying to run all html in that directory ### 5: Run `echo.sh` and compare with your predictions

```
bash scripts/echo.sh Oct07 Oct08
```

The output of the command was:

```
Oct07
Oct08
scripts/echo.sh: line 6: $3: unbound variable
```

```
bash scripts/echo.sh Oct07 Oct08 Oct09 Oct10
```

The output of the command was:

```
Oct07
Oct08
Oct09
Finished
```

```
bash scripts/echo.sh Oct07 Oct08 "Oct09 Oct10"
```

The output of the command was:

```
Oct07
Oct08
Oct09 Oct10
Finished
[reeves404@p0218
```

```
bash scripts/echo.sh results/fastqc/*.html
```

The output of the command was:

```
results/fastqc/S63_R1_fastqc.html
scripts/echo.sh: line 5: $2: unbound variable
```

Answer: Which output (if any) differed from your prediction, and why?

The second differed, as it ignored the fourth argument instead of error

6: Write the `printline.sh` script

7: Test `printline.sh` twice

Test 1:

```
# Your command: bash scripts/printline.sh 5 data/metadata.tsv
```

The output of the command was:

```
S83     10dpi   cathemerium
```

Test 2 (with redirection), and checking the contents of the output file:

```
# Your command: bash scripts/printline.sh 5 data/metadata.tsv > results/printlinetest.txt
```

The output of the command that checks the file contents was:

```
S83     10dpi   cathemerium
```

Part C: A script to run MultiQC

8: Get a MultiQC container link

Answer: The container link (URI) is: oras://community.wave.seqera.io/library/multiqc:1.35–562c547b518ebe20

9: MultiQC's help info, and the MultiQC command

Code to run `multiqc --help` with the container:

```
# Your command: apptainer exec oras://community.wave.seqera.io/library/multiqc:1.35--562c54
```

The MultiQC command to run MultiQC on `results/fastqc` with output in `results/multiqc`:

```
# Your command: apptainer exec oras://community.wave.seqera.io/library/multiqc:1.35--562c54
```

10: Write the `multiqc.sh` script

11: Run `multiqc.sh`

Code to run the script:

```
# Your command: bash scripts/multiqc.sh
```

Answer: Number of FastQC reports MultiQC found: 1 *Answer*: Does that match what you expected? Why/why not? Yes, there was only 1 listed

12: Make `multiqc.sh` accept arguments, and rerun it

Code to rerun the script:

```
# Your command:  bash scripts/multiqc.sh results/fastqc results/multiqc
```

13: Differences between the two MultiQC runs

Answer: Theres no differences in output, it just names the file something new because theyre duplicates

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Part D: Pandoc to render Markdown

14: The structure of the Pandoc command

Answer (options and arguments, and what each does): options - giving it an input and output . Arguments first is output (-o), second argument is input (md)

Answer (how does Pandoc know which output format to produce?): the extension of the output file

15: Render your README to PDF and HTML

```
pandoc -o README.pdf README.md
```

- Command to render to HTML:

```
# Your command:
```

16: Should the rendered files be committed?

Answer: Should the rendered files be committed to your Git repo? Why/why not?

17: Install the PDF Viewer extension and view the PDF

Answer: One observation about the rendered PDF file:

Part E: Publish your repo on GitHub

18: Create a GitHub repository

19: Connect your local repo to GitHub, and push

20: Create an Issue to tag the instructors

Bonus

21: Explore the MultiQC report

Answer (one insight you gained):

Answer (one thing you're confused about):

22: Pandoc versions

```
# Your command:
```

The output of the command(s) was:

Answer: Is the Pandoc version available by default the most recent one?

Answer: Which Pandoc version are you using after loading the module?